

COP2073C Module 8 Programming Assignment

Note: Be sure to review the course materials on ANOVA testing and view the lecture recording before attempting this assignment. ANOVA is covered at approximately 1h08m in the lecture recording, and a brief introduction to this assignment occurs at approximately 1h32m.

Instructions

Dr. Horrible has created 3 separate recipes for zombie cookies to test for their effectiveness at transforming unsuspecting party guests into brain-eating zombies. Recipe A contains flour, chocolate chips, sugar, eggs, and a teaspoon of secret ingredient X. Recipe B contains the same ingredients but with a tablespoon of secret ingredient X. Recipe C contains the same ingredients but with no secret ingredient X (the control). At his Halloween party, the doctor seats his guests at one of three tables labeled A, B, and C and includes a cookie with the associated recipe (A, B, C) for each guest. After the party, he observes the guests' behavior for symptoms of brain-eating and assigns a numeric transformation significance value based on these observations, where 0 is no brain-eating symptoms and 10 is indicative of an uncontrollable appetite for brains. His observed data is as follows:

Recipe A	Recipe B	Recipe C
0	7	1
4	6	0
4	4	1
5	2	0
4	3	NA
1	0	1
2	8	1
3	6	NA
3	2	1
5	3	0

Write an R script which performs the following tasks:

1. Create a data frame from the above table, then stack it using the stack function and remove the NA values with na.omit; save the results to a new data frame, e.g., assuming your data frame is named "df", use

```
newdf <- na.omit(stack(df))
```

3. Run a summary on the new data frame; save and observe the results; the response variable is the "values" column and the predictor is the "ind" column.
4. Run a one-way analysis of variance test on the saved data using the aov function with formula notation. Remember from the module reading material that formula notation looks like this:

aov(your response variable ~ your predictor variable, data=your dataframe)

5. In your script comments indicate the response and predictor and provide a null and alternative hypothesis.
6. Run another summary function with the aov data; save and observe the results.
7. Indicate if the null hypothesis can be rejected. You can extract the p-value using the following, assuming your data from step 6 is named "anova_summary", as follows:

pval <- anova_summary[[1]]\$`Pr(>F)`[1]

Non-Functional Requirements:

- Include a 4-line ID header at the beginning of your script.
- Include vertical spacing (a blank line) between logical blocks for readability.
- Comment your code thoughtfully (avoid excessive commenting).
- Ensure each line of code does not exceed 80 column

Expected Output:

stacked data frame:

	values	ind
1	0	RecipeA
2	4	RecipeA
3	4	RecipeA
4	5	RecipeA
5	4	RecipeA
6	1	RecipeA
7	2	RecipeA
8	3	RecipeA
9	3	RecipeA
10	5	RecipeA
11	7	RecipeB
12	6	RecipeB
13	4	RecipeB
14	2	RecipeB
15	3	RecipeB
16	0	RecipeB
17	8	RecipeB
18	6	RecipeB
19	2	RecipeB
20	3	RecipeB
21	1	RecipeC
22	0	RecipeC
23	1	RecipeC
24	0	RecipeC

26 1 RecipeC
27 1 RecipeC
29 1 RecipeC
30 0 RecipeC

initial summary:

	values	ind
Min.	:0.00	RecipeA:10
1st Qu.:	:1.00	RecipeB:10
Median	:2.50	RecipeC: 8
Mean	:2.75	
3rd Qu.:	:4.00	
Max.	:8.00	

anova summary:

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
ind	2	55.58	27.788	8.108	0.00193 **
Residuals	25	85.68	3.427		

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

p_value is 0.001931171

Sufficient evidence to reject the null hypothesis.
Ingredient X may have an effect on the results.